Social Media & Mental Health



By: Jared Lomeli (Group 13)

High Screen Time Doubles Depression and Anxiety Risk for Teens



According to the U.S. Department of Health and Human Services, Kids and teens who spend more than 3 hours a day on social media face **double the risk** of depression and anxiety symptoms.

Table of Contents

Usage and Screen Time [01]

The Mental Health Numbers [02]

The Perception Gap [03]

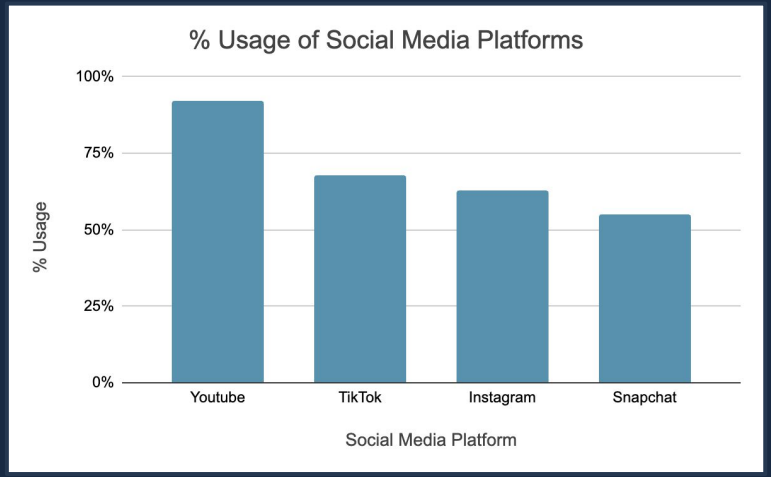
A Probability Lens on Risk [04]

Our Own Survey [05]

Takeaways and Sources [06]

01 Usage and Screen Time

How many people are actually on these apps?



1. Youtube

92%

2.TikTok

68%

3. Instagram

63%

4. Snapchat

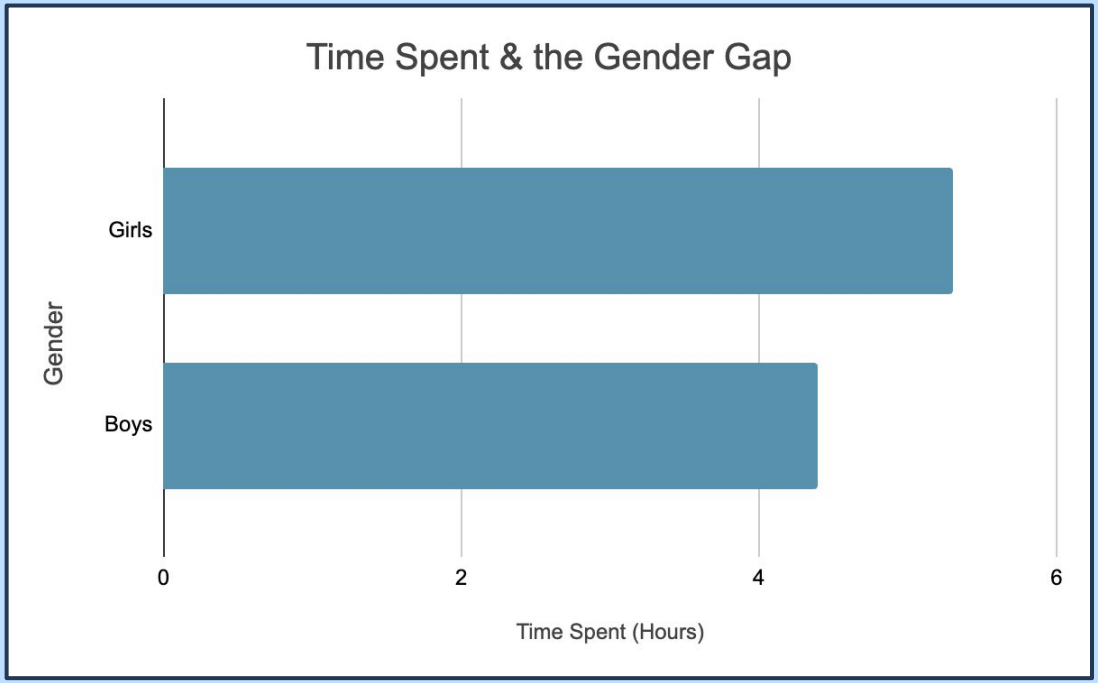
55%

Average Daily Use & the Gender Gap

According to a survey conducted between 2019 and 2022 showed that teenagers spend an average of 3.5 hours a day on social media.

2023 Survey: Average 4.8 hrs/day

- Girls average 5.3 hrs/day
- boys average 4.4 hrs/day.



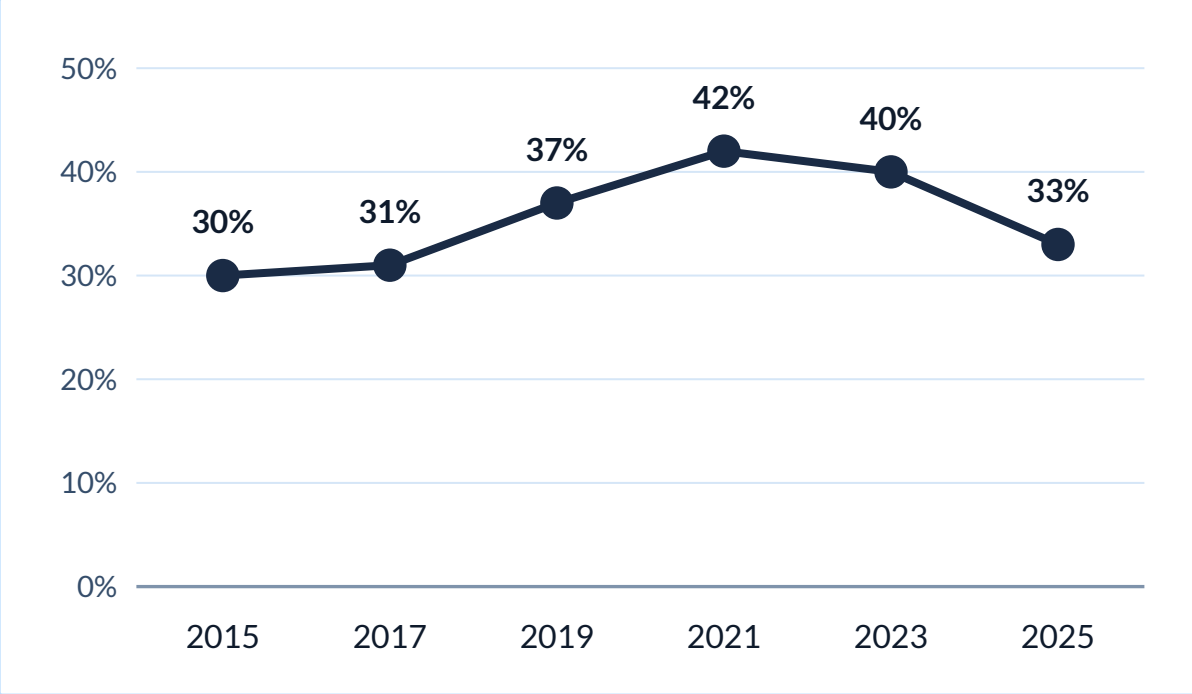
02

The Mental Health Numbers

What is happening to our mental health?

Teen sadness rose, then dropped

% of U.S. high schoolers who felt persistently sad or hopeless in the past year



42%

Peak in 2021

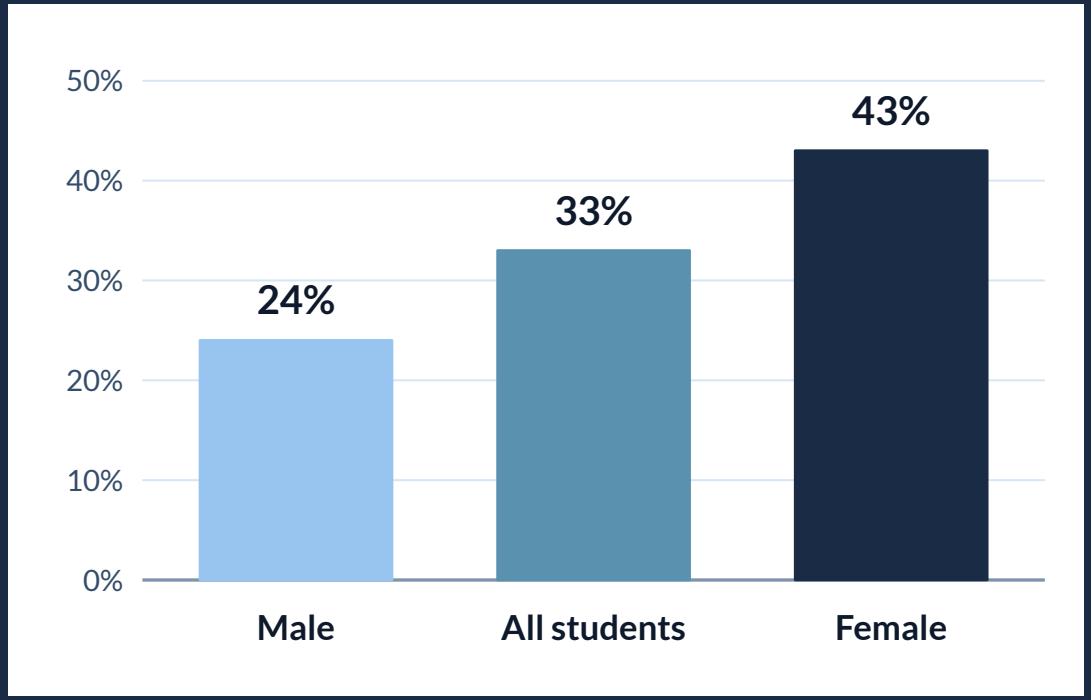
33%

Newest data, 2025

Source: CDC Youth Risk Behavior Survey, 2015–2025 (cdc.gov/yrbs)

Girls report it almost twice as often

Persistent sadness or hopelessness, U.S. high schoolers, 2025



1 in 3

high schoolers felt persistently sad or hopeless in 2025

Ch. 3 tie-in

One year, split by group = cross-sectional data, so a bar chart, not a line

Source: CDC Youth Risk Behavior Survey, 2025 (cdc.gov/yrbs)

Where social media comes in

What the U.S. Surgeon General found about heavy users

2x

the risk

Teens on social media 3+ hours a day face double the risk of depression and anxiety symptoms

46%

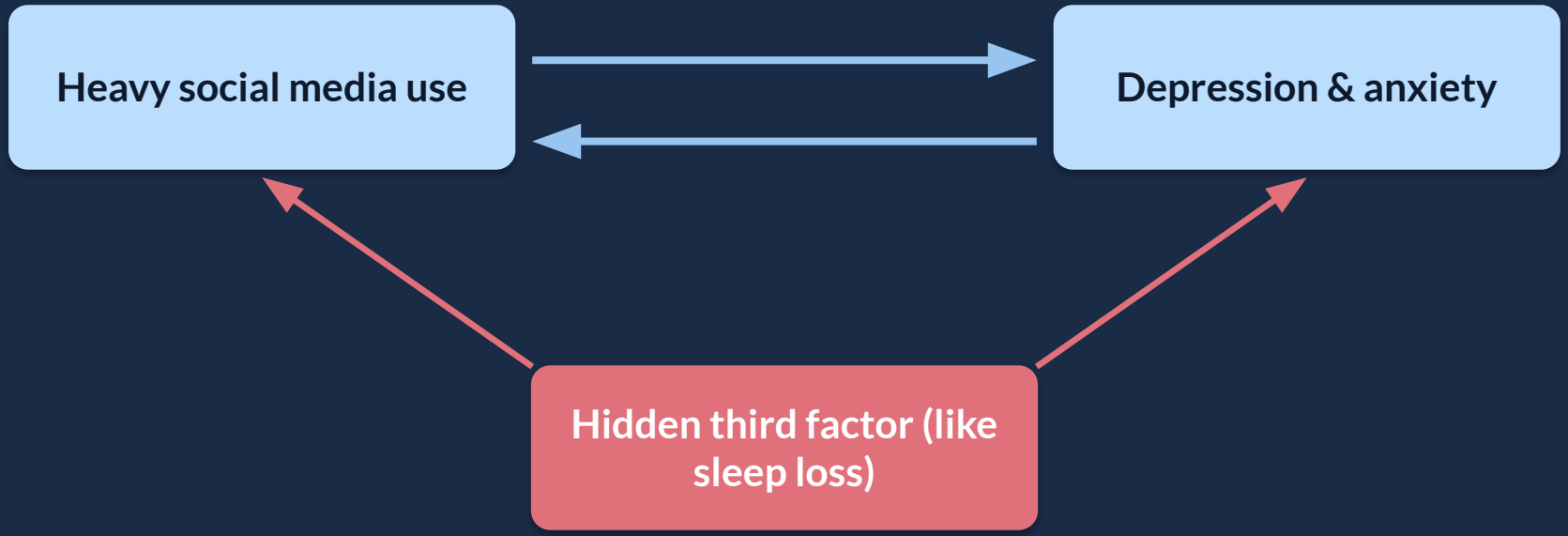
feel worse

of teens ages 13 to 17 say social media makes them feel worse about their body image

Source: U.S. Surgeon General Advisory, "Social Media and Youth Mental Health" (hhs.gov)

Linked is not the same as caused

Ch. 1, Statistical Pitfall #5: assuming a causal link



Could be driving both

03

The Perception Gap

The Gap

People think it’s bad for everyone else

48%

of teens say social media has a negative effect on people their age

14%

say it has a negative effect on themselves

Source: Pew Research Center, Teens, Social Media and Mental Health (Apr. 2025)

Positives vs Negatives

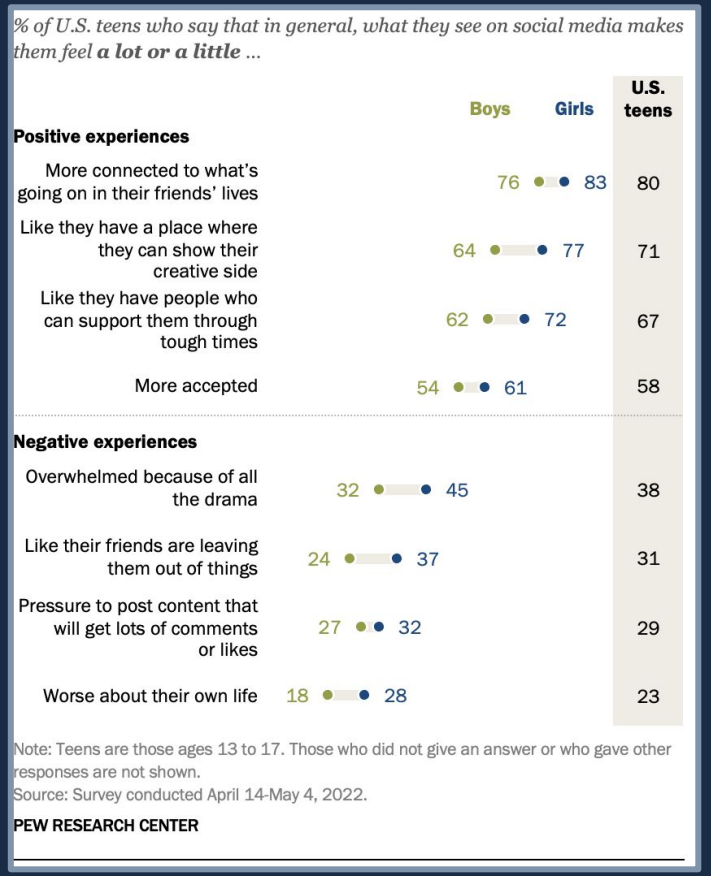
Persistent sadness or hopelessness, U.S. high schoolers, 2025

Positive

80% - feel more connected to friends
71% feel a creative outlet
67% feel supported through hard times
58% feel more accepted

Negatives

38% feel overwhelmed by “drama”
31% feel left out
29% feel pressure to post for likes
23% feel worse about their own life



Source: Pew Research Center, Teens and Social Media: Key Findings (2023)

04

A Probability Lens on Risk

What is the actual starting chance?

- Reframing the “2x the risk” as probability.
- Saying something “doubles your risk” sounds scary.
- What was the original risk to begin with?



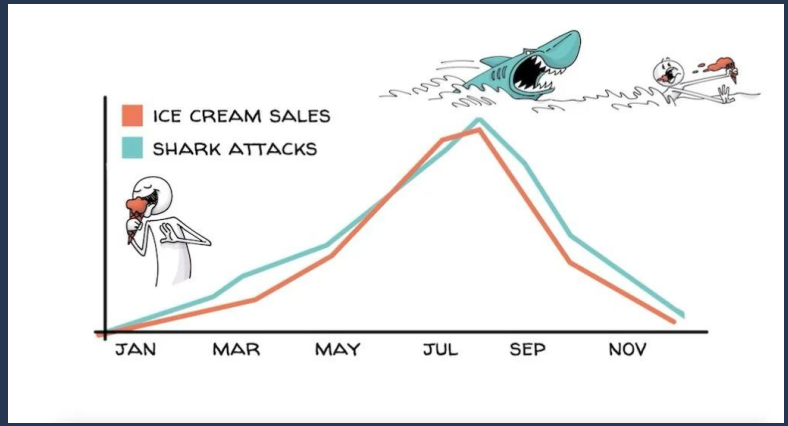
Who is really affected?

- Not everyone faces the same odds.
- Different Factors:
  - Gender
  - Existing Health
  - Offline Support



Does social media cause the harm?

- More scrolling, different reasons.
- Two things happening together doesn’t mean one caused the other.



05

Our Own Survey

Survey Methodology and Sampling Frame

Survey vs Poll

To obtain our data, we created a survey instead of a poll.

- Survey: multiple questions across wide range of question types
- Poll: single multiple choice questions

Target Population

SJSU undergrad students

Sampling Frame/Method

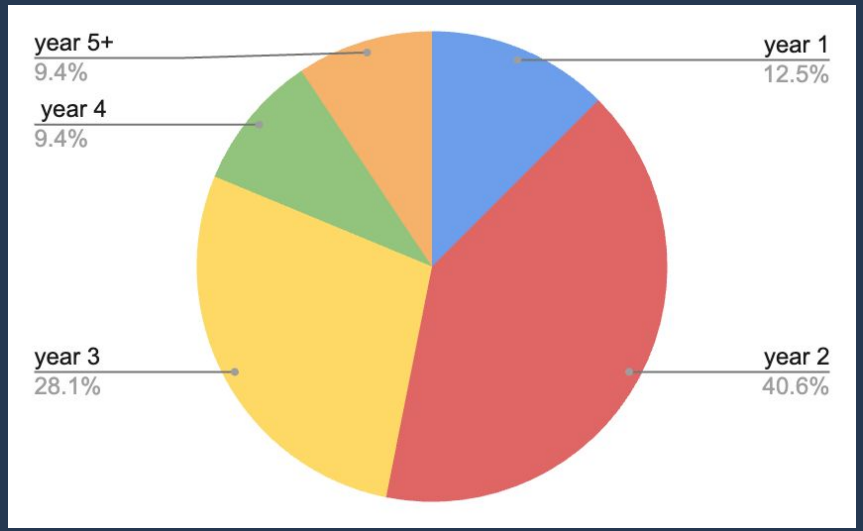
Respondents were made up of friends and classmates who who attend SJSU (**convenience sampling**: a non-random sampling relying on available respondents)

Methodological Limitations

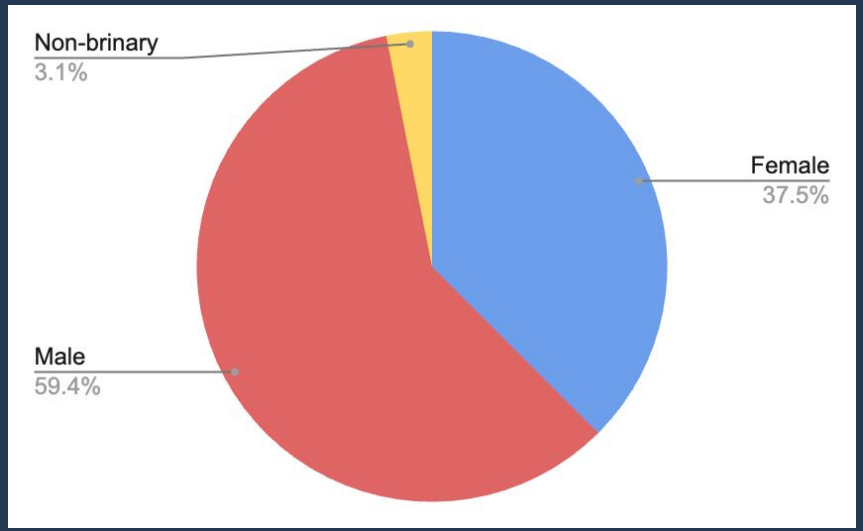
Pitfall #2 : convenience sampling is non-random, meaning out sample has voluntary response bias and cannot be generalized to the entire population (all of SJSU students)

Questionnaire Design and Results

Question 1: What year are you? (Categorical, Ordinal)



Question 2: What is your gender? (Categorical, Nominal)



Measures of Center

- Mean: 4.55 hours/day
- Median: 4 hours/day
- Mode: 4 hours/day

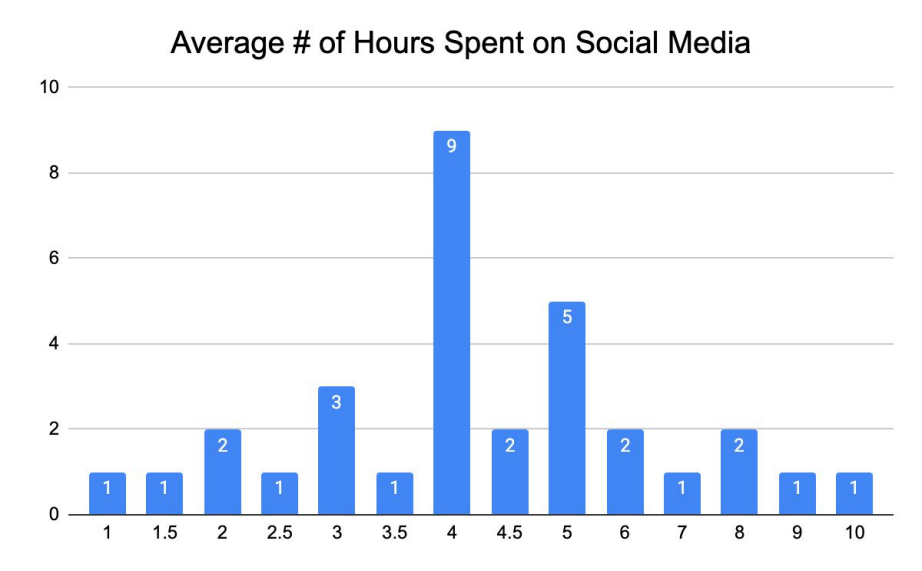
5-Number Summary & Spread

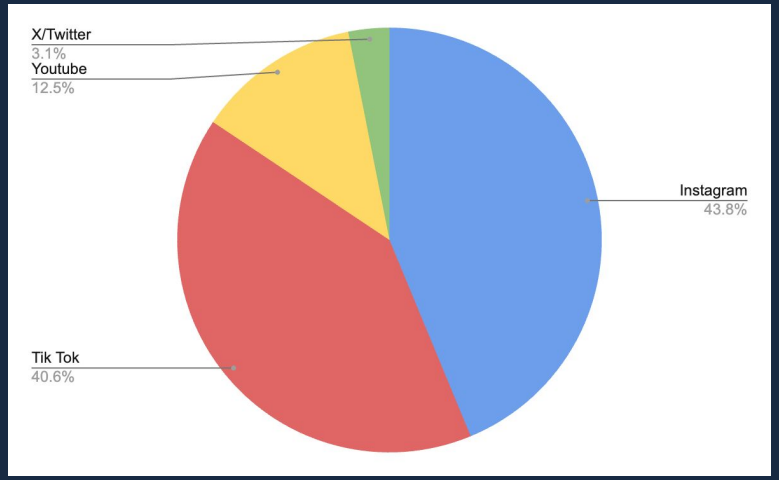
- Minimum: 1 hr
- Maximum: 10 hrs
- Q1: 3.25 hrs
- Q3: 5 hrs
- IQR: 1.75 hrs

Distribution Shape

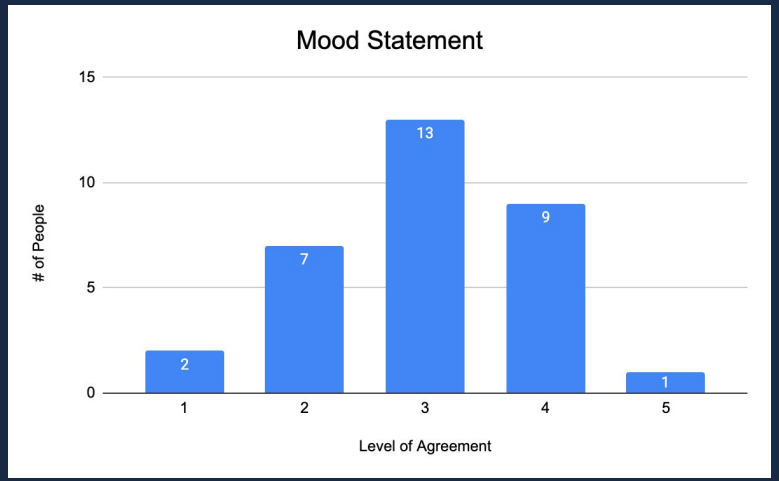
- Rightward Skewness
  Mean (4.55) > Median (4)

Question 3: On an average day, how many hours do you spend on social media? (Quantitative, Continuous)



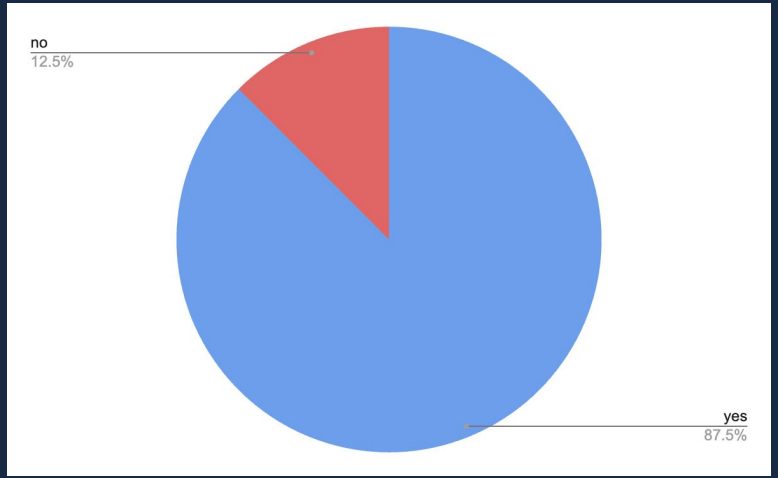


Question 4: Which social media platform do you use the most? (Categorical, Nominal)

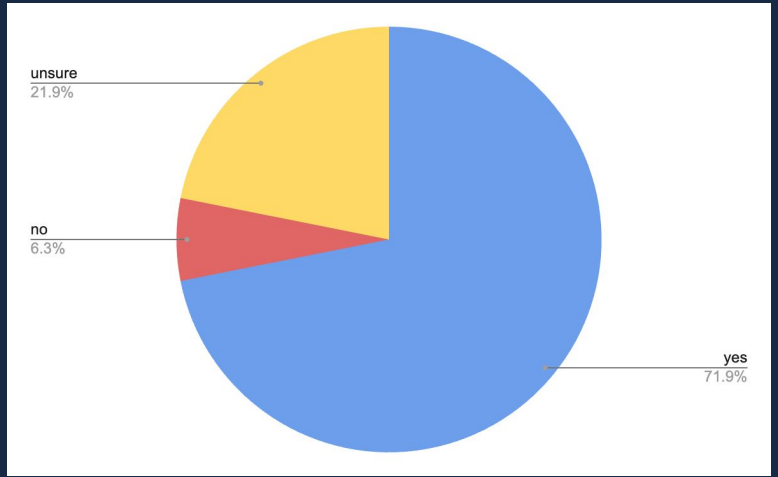


Question 5: Rate your agreement with the following statement: "Using social media generally improves my overall mood." (Categorical, Ordinal)

Question 6: Have you ever intentionally taken or tried to take a break or "detox" from social media? (Categorical, Nominal)



Question 7: Do you think social media affects your mental health? (Categorical, Nominal)



Frequency table

Usage Level	Mental Health Affected (Yes)	Mental Health Affected (No/Unsure)	Row Total
Heavy Use (> 3 hrs/day)	21	6	27 (84.4%)
Light Use (<3 hrs/day)	2	3	5 (15.6%)
Column Total	23 (71.9%)	9 (28.1%)	32 (100%)

Heavy Users: 21/27 = **77.8%** report mental health impact

Light Users: 2/5 = **40%** report mental health impact

06

Takeaways and Sources

Our Takeaways

Usage

- Social media is a constant presence in teens' lives
- Most spend hours on it every day

Mental Health

- Heavy use and worse mental health are linked

Perception Gap

- Teens see the downsides for others more than for themselves

Takeaways

Probability

- “Double the risk” means heavy users show symptoms about twice as often as light users
- A pattern across groups, not a prediction for everyone

Our Survey

- 84% of surveyed students spend 3+ hours daily on social media, and heavy users are twice as likely to report mental health impacts (77.8% vs. 40.0%)
- 87.5% of students have tried a digital detox, showing they actively recognize their heavy usage and want boundaries

What Can Be Done?

Set Boundaries

Create limits for the amount of time you spend online

Be mindful of what you see and share

Think before posting/ sharing

Don't follow content that makes you feel worse

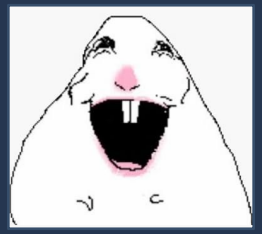
Reach Out

Talk to someone or report harassment is something online is harming you

Keep devices away before sleeping

Keep devices away from you before going to bed

Thank You



Works Cited

- Pew Research Center, *Teens, Social Media and AI Chatbots 2025* (Dec. 2025) — https://www.pewresearch.org/internet/2025/12/09/teens-social-media-and-ai-chatbots-2025/
- Pew Research Center, *Teens, Social Media and Mental Health* (Apr. 2025) — https://www.pewresearch.org/internet/2025/04/22/teens-social-media-and-mental-health/
- Pew Research Center, *Teens and Social Media: Key Findings* (2023) — https://www.pewresearch.org/internet/2023/04/24/teens-and-social-media-key-findings-from-pew-research-center-surveys/
- CDC, *Youth Risk Behavior Survey Data Summary & Trends Report* — https://www.cdc.gov/yrbs/dstr/index.html
- CDC Newsroom, *CDC Data Show Improvements in Youth Mental Health...* (2024) — https://www.cdc.gov/media/releases/2024/p0806-youth-mental-health.html
- ABC News/GMA, reporting 2023 CDC YRBS data on LGBTQ+ students — https://goodmorningamerica.com/news/story/40-lgbtq-youth-considered-suicide-past-year-cdc-112604907
- U.S. Surgeon General / HHS, *Social Media and Youth Mental Health* advisory — https://www.hhs.gov/surgeongeneral/reports-and-publications/youth-mental-health/social-media/index.html
- Riehm et al., 2019, *JAMA Psychiatry* (primary source behind the "2x risk" stat) — https://jamanetwork.com/journals/jamapsychiatry/fullarticle/2749480
- Bickham et al., 2022, Digital Wellness Lab (primary source behind the 46% body image stat) — https://digitalwellnesslab.org/wp-content/uploads/Pulse-Survey_Adolescent-Attitudes-Effects-and-Experiences.pdf
- Gallup, *Teens Spend Average of 4.8 Hours on Social Media Per Day* (2023) — https://news.gallup.com/poll/512576/teens-spend-average-hours-social-media-per-day.aspx
- https://explodingtopics.com/blog/screen-time-for-teens?utm_source